Variational Autoencoders

A Mathematical and Conceptual Study

Encoder, ELBO, Reparameterization Trick,

Gaussian Posteriors, KL Divergence, Posterior Collapse,
KL Annealing, Free Bits, and Mass-Covering Behavior

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Abstract

Variational Autoencoders (VAEs) are generative models based on latent-variable probabilistic models and variational inference. They provide a computationally efficient framework for jointly learning a generative model and an approximate inference model using stochastic gradient-based optimization.

This report develops the mathematical foundations of VAEs. It begins with the encoder or approximate posterior $q_\phi(z|x)$ and derives the Evidence Lower Bound (ELBO) as a lower bound on the marginal log-likelihood. The report then explains the relationship between the ELBO and Kullback–Leibler (KL) divergences, followed by the reparameterization trick that enables gradient-based optimization with continuous latent variables.

Factorized and full-covariance Gaussian posterior models are discussed, including the Cholesky parameterization and the computation of the Jacobian determinant. The report also examines posterior collapse and two common techniques for mitigating it: KL annealing and free bits.

Finally, the relationship between the direction of the KL divergence and the variance of the learned generative distribution is analyzed. In particular, the mass-covering behavior of $D_{\mathrm{KL}}(q\|p)$ is used to explain why a generative model may have a larger variance than the empirical data distribution when an exact fit is not possible.

Contents

1 Introduction

Variational Autoencoders (VAEs) are probabilistic generative models that combine deep neural networks with latent-variable modeling and variational inference. A VAE assumes that observed data x is generated through an unobserved latent variable z.

The generative model is commonly written as

$$p_\theta(x, z) = p_\theta(z)p_\theta(x|z), \tag{1}$$

where

- $p_\theta(z)$ is the prior distribution over latent variables,
- $p_\theta(x|z)$ is the decoder or likelihood model,
- $p_\theta(x, z)$ is the joint distribution,
- θ denotes the parameters of the generative model.

The marginal likelihood of an observation x is

$$p_\theta(x) = \int p_\theta(x, z)\, dz. \tag{2}$$

Although this expression is mathematically straightforward, the integral is generally intractable for high-dimensional latent variables.

The posterior distribution is

$$p_\theta(z|x) = \frac{p_\theta(x, z)}{p_\theta(x)}. \tag{3}$$

Because evaluating $p_\theta(x)$ is generally difficult, exact posterior inference is also difficult.

VAEs solve this problem by introducing an approximate posterior

$$q_\phi(z|x), \tag{4}$$

which is implemented using an encoder neural network.

Thus, a VAE contains two important mappings:

$$x \longrightarrow q_\phi(z|x) \longrightarrow z \longrightarrow p_\theta(x|z). \tag{5}$$

The encoder maps observations to a distribution over latent variables, while the decoder maps latent variables back to the distribution of observations.

2 Encoder and Approximate Posterior

The inference model of a VAE is denoted by

$$q_\phi(z|x). \tag{6}$$

It is also called the encoder or recognition model.

The objective of the encoder is to approximate the true posterior:

$$\boxed{q_\phi(z|x) \approx p_\theta(z|x)} \tag{7}$$

where ϕ represents the variational parameters.

The encoder can be implemented using a neural network. For example,

$$(\mu, \log \sigma) = \text{EncoderNN}_\phi(x). \tag{8}$$

The encoder then defines a Gaussian approximate posterior:

$$q_\phi(z|x) = \mathcal{N}\left(z; \mu, \text{diag}(\sigma^2)\right). \tag{9}$$

The use of $\log \sigma$ rather than σ directly is common because it allows the network to produce unconstrained real-valued outputs while ensuring that the standard deviation can be obtained through

$$\sigma_i = e^{\log \sigma_i}. \tag{10}$$

3 Amortized Variational Inference

Traditional variational inference may optimize a separate set of variational parameters for every datapoint.

VAEs instead use a single encoder network whose parameters are shared across all datapoints.

For each datapoint $x^{(i)}$,

$$q_\phi(z|x^{(i)}) \tag{11}$$

is produced by the same neural network.

Thus,

$$(\mu^{(i)}, \log \sigma^{(i)}) = \text{EncoderNN}_\phi(x^{(i)}). \tag{12}$$

This approach is known as *amortized variational inference*.

The advantage is that we avoid solving a separate optimization problem for every datapoint. Instead, the shared parameters ϕ are learned using stochastic gradient descent.

Therefore,

$$\boxed{\text{One encoder} \quad \Longrightarrow \quad \text{inference for many datapoints}} \tag{13}$$

This makes variational inference computationally efficient for large datasets.

4 Evidence Lower Bound

4.1 Derivation of the ELBO

The marginal likelihood is

$$p_\theta(x) = \int p_\theta(x, z)\, dz. \tag{14}$$

Introduce the approximate posterior $q_\phi(z|x)$:

$$p_\theta(x) = \int q_\phi(z|x) \frac{p_\theta(x, z)}{q_\phi(z|x)}\, dz. \tag{15}$$

Therefore,

$$p_\theta(x) = \mathbb{E}_{q_\phi(z|x)} \left[\frac{p_\theta(x, z)}{q_\phi(z|x)} \right]. \tag{16}$$

Taking the logarithm,

$$\log p_\theta(x) = \log \mathbb{E}_{q_\phi(z|x)} \left[\frac{p_\theta(x, z)}{q_\phi(z|x)} \right]. \tag{17}$$

Using Jensen's inequality,

$$\log \mathbb{E}[Y] \geq \mathbb{E}[\log Y], \tag{18}$$

we obtain

$$\log p_\theta(x) \geq \mathbb{E}_{q_\phi(z|x)} \left[\log \frac{p_\theta(x, z)}{q_\phi(z|x)} \right]. \tag{19}$$

The right-hand side is the Evidence Lower Bound:

$$\boxed{\mathcal{L}_{\theta,\phi}(x) = \mathbb{E}_{q_\phi(z|x)} \left[\log p_\theta(x, z) - \log q_\phi(z|x)\right]} \tag{20}$$

Hence,

$$\boxed{\mathcal{L}_{\theta,\phi}(x) \leq \log p_\theta(x)} \tag{21}$$

which explains the name *Evidence Lower Bound*.

4.2 ELBO as a KL Decomposition

Using Bayes' rule,

$$p_\theta(z|x) = \frac{p_\theta(x, z)}{p_\theta(x)}. \tag{22}$$

Therefore,

$$\log p_\theta(x, z) = \log p_\theta(z|x) + \log p_\theta(x). \tag{23}$$

Substituting into the ELBO gives

$$\begin{aligned} \mathcal{L}_{\theta,\phi}(x) &= \mathbb{E}_{q_\phi(z|x)} \left[\log p_\theta(z|x) + \log p_\theta(x) - \log q_\phi(z|x)\right] \\ &= \log p_\theta(x) - \mathbb{E}_{q_\phi(z|x)} \left[\log \frac{q_\phi(z|x)}{p_\theta(z|x)}\right]. \end{aligned} \tag{24}$$

The second term is the KL divergence:

$$D_{\mathrm{KL}} \left(q_\phi(z|x) \| p_\theta(z|x)\right). \tag{25}$$

Hence,

$$\boxed{\mathcal{L}_{\theta,\phi}(x) = \log p_\theta(x) - D_{\mathrm{KL}} \left(q_\phi(z|x) \| p_\theta(z|x)\right)} \tag{26}$$

Since

$$D_{\mathrm{KL}}(q \| p) \geq 0, \tag{27}$$

we obtain

$$\boxed{\mathcal{L}_{\theta,\phi}(x) \leq \log p_\theta(x)} \tag{28}$$

The difference between the marginal log-likelihood and the ELBO is

$$\boxed{\log p_\theta(x) - \mathcal{L}_{\theta,\phi}(x) = D_{\mathrm{KL}} \left(q_\phi(z|x) \| p_\theta(z|x)\right)} \tag{29}$$

Thus, the KL divergence determines the tightness of the ELBO.

5 ELBO for a Dataset

Consider a dataset

$$D = \left\{x^{(1)}, x^{(2)}, \ldots, x^{(N_D)}\right\}. \tag{30}$$

The average dataset ELBO is

$$\mathcal{L}_{\theta,\phi}(D) = \frac{1}{N_D} \sum_{i=1}^{N_D} \mathcal{L}_{\theta,\phi}\left(x^{(i)}\right). \tag{31}$$

Substituting the individual ELBO,

$$\mathcal{L}_{\theta,\phi}(D) = \frac{1}{N_D} \sum_{i=1}^{N_D} \mathbb{E}_{q_\phi(z|x^{(i)})}\left[\log p_\theta(x^{(i)}, z) - \log q_\phi(z|x^{(i)})\right]. \tag{32}$$

Define the empirical data distribution as

$$\boxed{q_D(x) = \frac{1}{N_D} \sum_{i=1}^{N_D} \delta(x - x^{(i)})} \tag{33}$$

for continuous data.

For any function $f(x)$,

$$\boxed{\mathbb{E}_{q_D(x)}[f(x)] = \frac{1}{N_D} \sum_{i=1}^{N_D} f(x^{(i)})} \tag{34}$$

Therefore,

$$\boxed{\mathcal{L}_{\theta,\phi}(D) = \mathbb{E}_{q_D(x)}\left[\mathbb{E}_{q_\phi(z|x)}\left[\log p_\theta(x, z) - \log q_\phi(z|x)\right]\right]} \tag{35}$$

Thus, the expectation over $q_D(x)$ is simply another way of writing the average over the dataset.

6 Reparameterization Trick

Directly sampling

$$z \sim q_\phi(z|x) \tag{36}$$

creates a difficulty when computing gradients with respect to ϕ, because the sampling operation itself depends on ϕ.

For continuous latent variables, this difficulty can be addressed using the reparameterization trick.

We express z as

$$\boxed{z = g(\epsilon, \phi, x)} \tag{37}$$

where

$$\epsilon \sim p(\epsilon) \tag{38}$$

and the distribution of ϵ does not depend on ϕ or x.

For a Gaussian posterior,

$$q_\phi(z|x) = \mathcal{N}\left(z; \mu, \mathrm{diag}(\sigma^2)\right), \tag{39}$$

we use

$$\boxed{\epsilon \sim \mathcal{N}(0, I)} \tag{40}$$

and

$$\boxed{z = \mu + \sigma \odot \epsilon} \tag{41}$$

where $\odot$ denotes element-wise multiplication.

Now the randomness is entirely contained in ϵ, while z is a deterministic and differentiable function of μ, σ, and ϵ.

Consequently, gradients can be propagated through

$$z = \mu + \sigma \odot \epsilon. \tag{42}$$

7 Stochastic Gradient Optimization

The ELBO for one datapoint can be written as

$$\mathcal{L}_{\theta,\phi}(x) = \mathbb{E}_{q_\phi(z|x)}\left[\log p_\theta(x, z) - \log q_\phi(z|x)\right]. \tag{43}$$

After reparameterization,

$$\mathcal{L}_{\theta,\phi}(x) = \mathbb{E}_{p(\epsilon)}\left[\log p_\theta(x, z) - \log q_\phi(z|x)\right], \tag{44}$$

where

$$z = g(\epsilon, \phi, x). \tag{45}$$

A Monte Carlo estimate using a single sample is

$$\boxed{\widetilde{\mathcal{L}}_{\theta,\phi}(x) = \log p_\theta(x,z) - \log q_\phi(z|x)} \tag{46}$$

where

$$\epsilon \sim p(\epsilon), \qquad z = g(\epsilon, \phi, x). \tag{47}$$

The gradient can then be computed using ordinary backpropagation:

$$\nabla_{\theta,\phi}\widetilde{\mathcal{L}}_{\theta,\phi}(x). \tag{48}$$

A minibatch provides a stochastic estimate of the full dataset gradient.

8 Change of Variables and Log-Density

Suppose

$$z = g(\epsilon, \phi, x) \tag{49}$$

is an invertible transformation.

The densities of ϵ and z are related by the change-of- variables formula:

$$q_\phi(z|x) = p(\epsilon)\left|\det\left(\frac{\partial z}{\partial \epsilon}\right)\right|^{-1}. \tag{50}$$

Taking logarithms gives

$$\boxed{\log q_\phi(z|x) = \log p(\epsilon) - \log\left|\det\left(\frac{\partial z}{\partial \epsilon}\right)\right|} \tag{51}$$

The Jacobian matrix is

$$\frac{\partial z}{\partial \epsilon} = \begin{pmatrix} \frac{\partial z_1}{\partial \epsilon_1} & \cdots & \frac{\partial z_1}{\partial \epsilon_k} \\ \vdots & \ddots & \vdots \\ \frac{\partial z_k}{\partial \epsilon_1} & \cdots & \frac{\partial z_k}{\partial \epsilon_k} \end{pmatrix}. \tag{52}$$

9 Factorized Gaussian Posterior

A common choice for the approximate posterior is a factorized Gaussian:

$$\boxed{q_\phi(z|x) = \prod_{i=1}^{d} \mathcal{N}(z_i; \mu_i, \sigma_i^2)} \tag{53}$$

Equivalently,

$$q_\phi(z|x) = \mathcal{N}\left(z; \mu, \mathrm{diag}(\sigma^2)\right). \tag{54}$$

The reparameterization is

$$\epsilon \sim \mathcal{N}(0, I), \tag{55}$$

$$\boxed{z = \mu + \sigma \odot \epsilon.} \tag{56}$$

The Jacobian is

$$\frac{\partial z}{\partial \epsilon} = \mathrm{diag}(\sigma). \tag{57}$$

Therefore,

$$\det\left(\frac{\partial z}{\partial \epsilon}\right) = \prod_i \sigma_i. \tag{58}$$

Hence,

$$\boxed{\log\left|\det\left(\frac{\partial z}{\partial \epsilon}\right)\right| = \sum_i \log|\sigma_i|} \tag{59}$$

and therefore

$$\boxed{\log q_\phi(z|x) = \log p(\epsilon) - \sum_i \log|\sigma_i|} \tag{60}$$

For $\epsilon \sim \mathcal{N}(0, I)$,

$$\log p(\epsilon) = -\frac{1}{2}\sum_i \left(\epsilon_i^2 + \log(2\pi)\right). \tag{61}$$

Thus,

$$\boxed{\log q_\phi(z|x) = -\frac{1}{2}\sum_i \left(\epsilon_i^2 + \log(2\pi)\right) - \sum_i \log|\sigma_i|} \tag{62}$$

10 Full-Covariance Gaussian Posterior

The factorized Gaussian assumes that the latent dimensions are conditionally independent given x.

A more flexible posterior is

$$\boxed{q_\phi(z|x) = \mathcal{N}(z; \mu, \Sigma)} \tag{63}$$

where Σ is a full covariance matrix.

We can parameterize z using

$$\epsilon \sim \mathcal{N}(0, I) \tag{64}$$

and

$$\boxed{z = \mu + L\epsilon} \tag{65}$$

where L is a triangular matrix.

The Jacobian is simply

$$\frac{\partial z}{\partial \epsilon} = L. \tag{66}$$

Because L is triangular,

$$\det(L) = \prod_i L_{ii}. \tag{67}$$

Therefore,

$$\boxed{\log|\det(L)| = \sum_i \log|L_{ii}|} \tag{68}$$

and

$$\boxed{\log q_\phi(z|x) = \log p(\epsilon) - \sum_i \log|L_{ii}|} \tag{69}$$

10.1 Cholesky Decomposition

The covariance matrix of z is

$$\Sigma = \mathrm{Cov}(z). \tag{70}$$

Since

$$z = \mu + L\epsilon \tag{71}$$

and

$$\mathbb{E}[\epsilon] = 0, \qquad \mathbb{E}[\epsilon\epsilon^T] = I, \tag{72}$$

we have

$$
\begin{aligned}
\Sigma &= \mathbb{E}\left[(z-\mu)(z-\mu)^T\right] \\
&= \mathbb{E}\left[L\epsilon\epsilon^T L^T\right] \\
&= L\mathbb{E}\left[\epsilon\epsilon^T\right] L^T \\
&= LL^T.
\end{aligned}
\tag{73}
$$

Thus,

$$
\boxed{\Sigma = LL^T} \tag{74}
$$

which is the Cholesky decomposition.

10.2 Constructing the Triangular Matrix

A neural network can output

$$
(\mu, \log\sigma, L') = \mathrm{EncoderNN}_\phi(x). \tag{75}
$$

A lower-triangular matrix can be constructed using a masking matrix:

$$
\boxed{L = L_{\mathrm{mask}} L' + \mathrm{diag}(\sigma)} \tag{76}
$$

where L_{mask} contains ones below the diagonal and zeros elsewhere.

The diagonal entries are determined by σ, ensuring that the Jacobian determinant remains easy to compute.

11 Marginal Likelihood and ELBO as KL Divergences

The empirical data distribution can be written as

$$
q_D(x) = \frac{1}{N_D}\sum_{i=1}^{N_D}\delta(x - x^{(i)}). \tag{77}
$$

The maximum-likelihood objective is

$$
\frac{1}{N_D}\sum_{i=1}^{N_D}\log p_\theta(x^{(i)}). \tag{78}
$$

Equivalently,

$$
\log p_\theta(D) = \mathbb{E}_{q_D(x)}[\log p_\theta(x)]. \tag{79}
$$

The KL divergence between the empirical data distribution and the model distribution is

$$
\begin{aligned}
D_{\mathrm{KL}}\left(q_D(x)\|p_\theta(x)\right) &= \mathbb{E}_{q_D(x)}\left[\log\frac{q_D(x)}{p_\theta(x)}\right] \\
&= \mathbb{E}_{q_D(x)}[\log q_D(x)] - \mathbb{E}_{q_D(x)}[\log p_\theta(x)].
\end{aligned} \tag{80}
$$

Since the first term is independent of θ,

$$
\boxed{D_{\mathrm{KL}}\left(q_D(x)\|p_\theta(x)\right) = -\log p_\theta(D) + \text{constant}} \tag{81}
$$

Therefore,

$$
\boxed{\text{Maximum likelihood} \quad \Longleftrightarrow \quad \text{Minimization of } D_{\mathrm{KL}}(q_D(x)\|p_\theta(x)).} \tag{82}
$$

12 Joint KL and the ELBO

Define

$$
q_{D,\phi}(x,z) = q_D(x)q_\phi(z|x). \tag{83}
$$

The joint KL divergence is

$$
D_{\mathrm{KL}}\left(q_{D,\phi}(x,z)\|p_\theta(x,z)\right). \tag{84}
$$

Using the KL decomposition,

$$
\boxed{\begin{aligned}
D_{\mathrm{KL}}\left(q_{D,\phi}(x,z)\|p_\theta(x,z)\right) &= D_{\mathrm{KL}}\left(q_D(x)\|p_\theta(x)\right) \\
&\quad + \mathbb{E}_{q_D(x)}\left[D_{\mathrm{KL}}\left(q_\phi(z|x)\|p_\theta(z|x)\right)\right].
\end{aligned}} \tag{85}
$$

Since the second term is non-negative,

$$
\boxed{D_{\mathrm{KL}}\left(q_{D,\phi}(x,z)\|p_\theta(x,z)\right) \geq D_{\mathrm{KL}}\left(q_D(x)\|p_\theta(x)\right)} \tag{86}
$$

The ELBO is therefore equivalent, up to a constant, to minimizing the joint KL divergence:

$$
\boxed{D_{\mathrm{KL}}\left(q_{D,\phi}(x,z)\|p_\theta(x,z)\right) = -\mathcal{L}_{\theta,\phi}(D) + \text{constant}} \tag{87}
$$

Thus, maximizing the ELBO is equivalent to minimizing

$$
D_{\mathrm{KL}}\left(q_{D,\phi}(x,z)\|p_\theta(x,z)\right). \tag{88}
$$

13 Posterior Collapse

Posterior collapse occurs when the approximate posterior becomes close to the prior:

$$\boxed{q_\phi(z|x) \approx p(z)} \tag{89}$$

for many or all observations x.

The standard ELBO can be written as

$$\boxed{\mathcal{L} = \underbrace{\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)]}_{\text{Reconstruction}} - \underbrace{D_{\mathrm{KL}}(q_\phi(z|x)\|p(z))}_{\text{KL cost}}} \tag{90}$$

If

$$q_\phi(z|x) \approx p(z), \tag{91}$$

then

$$D_{\mathrm{KL}}(q_\phi(z|x)\|p(z)) \approx 0. \tag{92}$$

However, because $q_\phi(z|x)$ is then approximately independent of x, the latent variable contains little information about the input.

Hence,

$$\boxed{q_\phi(z|x) \approx p(z) \quad \Longrightarrow \quad I_q(x;z) \approx 0} \tag{93}$$

where $I_q(x;z)$ denotes the mutual information between x and z under the joint distribution induced by $q_D(x)q_\phi(z|x)$.

The optimization can therefore converge to an undesirable solution where the decoder largely ignores the latent representation.

14 KL Annealing

To reduce posterior collapse, the KL term can be gradually introduced during training.

The modified objective is

$$\boxed{\mathcal{L}_\beta = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \beta D_{\mathrm{KL}}(q_\phi(z|x)\|p(z))} \tag{94}$$

where

$$0 \leq \beta \leq 1. \tag{95}$$

Initially,

$$\beta \approx 0. \tag{96}$$

Therefore,

$$\mathcal{L}_\beta \approx \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)]. \tag{97}$$

The model is first encouraged to learn a useful representation.

Then β is gradually increased:

$$\boxed{0 \longrightarrow 1} \tag{98}$$

At the end of training,

$$\beta = 1, \tag{99}$$

and the standard ELBO is recovered.

15 Free Bits

Another approach is the free-bits method.

Divide the latent variables into K groups:

$$z = (z_1, z_2, \ldots, z_K). \tag{100}$$

Define the KL contribution of group j as

$$\mathcal{K}_j = \mathbb{E}_{x \sim M}\left[D_{\mathrm{KL}}\left(q_\phi(z_j|x) \| p(z_j)\right)\right]. \tag{101}$$

A threshold λ is introduced, and the KL contribution is modified using

$$\boxed{\max(\lambda, \mathcal{K}_j)} \tag{102}$$

The modified objective can therefore be expressed as

$$\boxed{\mathcal{L}^\lambda = \mathbb{E}_{x \sim M}\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \sum_{j=1}^{K} \max(\lambda, \mathcal{K}_j)} \tag{103}$$

If

$$\mathcal{K}_j < \lambda, \tag{104}$$

then the KL contribution is effectively treated as λ rather than being increased according to its smaller value.

For example,

$$\lambda = 0.5, \qquad \mathcal{K}_j = 0.2 \tag{105}$$

gives

$$\max(0.5, 0.2) = 0.5. \tag{106}$$

If instead

$$\mathcal{K}_j = 0.8, \tag{107}$$

then

$$\max(0.5, 0.8) = 0.8. \tag{108}$$

The purpose is to allow the latent variables to encode a minimum amount of information without immediately receiving the full optimization pressure toward the prior.

16 Direction of KL Divergence

The direction of KL divergence is important.

Consider

$$\boxed{D_{\mathrm{KL}}(q\|p) = \mathbb{E}_q\left[\log \frac{q}{p}\right].} \tag{109}$$

Equivalently,

$$D_{\mathrm{KL}}(q\|p) = \mathbb{E}_q[\log q] - \mathbb{E}_q[\log p]. \tag{110}$$

The expectation is taken with respect to q.

Suppose

$$q(x) > 0 \tag{111}$$

but

$$p(x) \approx 0. \tag{112}$$

Then

$$\log p(x) \to -\infty, \tag{113}$$

so

$$-\log p(x) \to +\infty. \tag{114}$$

Therefore,

$$\boxed{q(x) > 0, \quad p(x) \approx 0 \quad \Longrightarrow \quad D_{\mathrm{KL}}(q\|p) \text{ is heavily penalized.}} \tag{115}$$

Thus, the model is encouraged to assign probability to regions where the data distribution has probability.

17 Mass-Covering Behavior

The forward KL divergence

$$D_{\mathrm{KL}}(q\|p) \tag{116}$$

is often described as having a mass-covering tendency.

Suppose q has two high-probability regions A and B.

If p covers only A,

$$p(x) \approx 0 \qquad \text{for } x \in B, \tag{117}$$

then

$$q(x) > 0 \qquad \text{for } x \in B \tag{118}$$

causes a large contribution to

$$D_{\mathrm{KL}}(q\|p). \tag{119}$$

Therefore, the model is encouraged to cover both regions.

In contrast, if

$$p(x) > 0 \tag{120}$$

in a region where

$$q(x) \approx 0, \tag{121}$$

the contribution to $D_{\mathrm{KL}}(q\|p)$ is relatively small because the expectation is taken under q.

Hence,

$$\boxed{D_{\mathrm{KL}}(q\|p) \text{ strongly penalizes missing mass}} \tag{122}$$

while

$$\boxed{D_{\mathrm{KL}}(q\|p) \text{ is relatively tolerant of extra mass}} \tag{123}$$

This is the basic reason for the mass-covering behavior.

18 Why the Generative Distribution Can Have Higher Variance

Suppose the true data distribution cannot be represented exactly by the model family.

Consider the data distribution

$$q(x) = \frac{1}{2}\mathcal{N}(-2, 0.1^2) + \frac{1}{2}\mathcal{N}(2, 0.1^2). \tag{124}$$

This distribution contains two narrow modes.

Suppose the model can only represent the data using one Gaussian:

$$p_\theta(x) = \mathcal{N}(\mu, \sigma^2). \tag{125}$$

A narrow Gaussian centered around -2 would fail to assign sufficient probability to the second mode around 2.

Thus,

$$q(2) > 0, \qquad p_\theta(2) \approx 0, \tag{126}$$

which produces a large forward-KL penalty.

A broader Gaussian centered near the middle of the two modes can instead assign non-negligible probability to both regions:

$$p_\theta(x) = \mathcal{N}(0, \sigma^2), \tag{127}$$

with sufficiently large σ^2.

Although this model also assigns probability to regions where $q(x)$ has little probability, those regions are not strongly penalized by $D_{\mathrm{KL}}(q\|p)$.

Therefore, when an exact fit is impossible, the model may choose a broader distribution.

This can result in

$$\boxed{\mathrm{Var}[p_\theta(x)] > \mathrm{Var}[q(x)]} \tag{128}$$

and similarly, in the joint space,

$$\boxed{\mathrm{Var}[p_\theta(x,z)] > \mathrm{Var}[q_{D,\phi}(x,z)]} \tag{129}$$

in situations where the model cannot perfectly represent the empirical distribution.

19 Forward KL versus Reverse KL

The forward KL is

$$D_{\mathrm{KL}}(q\|p) = \mathbb{E}_q\left[\log \frac{q}{p}\right]. \tag{130}$$

Because the expectation is under q, missing a region where q has significant probability is heavily penalized.

Thus,

$$\boxed{D_{\mathrm{KL}}(q\|p) \quad \text{has a mass-covering tendency.}} \tag{131}$$

The reverse KL is

$$D_{\mathrm{KL}}(p\|q) = \mathbb{E}_p\left[\log \frac{p}{q}\right]. \tag{132}$$

Now the expectation is under p.

Consequently, if the model assigns probability to regions where

$$q(x) \approx 0, \tag{133}$$

the divergence becomes very large.

Thus, the reverse KL tends to discourage the model from assigning mass to unsupported regions and can exhibit a more mode-seeking behavior.

The distinction can be summarized as

KL direction	Typical behavior
$D_{\mathrm{KL}}(q\|p)$	Mass covering
$D_{\mathrm{KL}}(p\|q)$	More mode seeking

(134)

20 Overall VAE Objective

The standard VAE objective for a datapoint is

$$\boxed{\mathcal{L}_{\theta,\phi}(x) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - D_{\mathrm{KL}}(q_\phi(z|x)\|p(z))} \tag{135}$$

The first term encourages accurate reconstruction:

$$\boxed{\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)]} \tag{136}$$

while the second term regularizes the latent distribution:

$$\boxed{D_{\mathrm{KL}}(q_\phi(z|x)\|p(z))} \tag{137}$$

The optimization therefore balances

$$\boxed{\text{Reconstruction quality} \quad \text{vs.} \quad \text{Latent regularization.}} \tag{138}$$

The reparameterization trick makes it possible to optimize both θ and ϕ using stochastic gradient descent.

21 Complete Conceptual Picture

The complete mathematical picture can be summarized as follows.

The encoder produces an approximate posterior:

$$\boxed{q_\phi(z|x) \approx p_\theta(z|x)} \tag{139}$$

The ELBO is

$$\boxed{\mathcal{L}_{\theta,\phi}(x) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x,z) - \log q_\phi(z|x)]} \tag{140}$$

and satisfies

$$\boxed{\mathcal{L}_{\theta,\phi}(x) \leq \log p_\theta(x).} \tag{141}$$

The difference is

$$\boxed{\log p_\theta(x) - \mathcal{L}_{\theta,\phi}(x) = D_{\mathrm{KL}}(q_\phi(z|x)\|p_\theta(z|x))} \tag{142}$$

The dataset ELBO can be written using the empirical data distribution:

$$\boxed{\mathcal{L}_{\theta,\phi}(D) = \mathbb{E}_{q_D(x)}\left[\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x,z) - \log q_\phi(z|x)]\right]} \tag{143}$$

The corresponding joint KL is

$$\boxed{\begin{aligned} D_{\mathrm{KL}}(q_{D,\phi}(x,z)\|p_\theta(x,z)) = D_{\mathrm{KL}}(q_D(x)\|p_\theta(x)) \\ + \mathbb{E}_{q_D(x)}[D_{\mathrm{KL}}(q_\phi(z|x)\|p_\theta(z|x))]. \end{aligned}} \tag{144}$$

Because the second term is non-negative,

$$\boxed{D_{\mathrm{KL}}(q_{D,\phi}(x,z)\|p_\theta(x,z)) \geq D_{\mathrm{KL}}(q_D(x)\|p_\theta(x)).} \tag{145}$$

For continuous latent variables, the reparameterization trick is

$$\boxed{z = g(\epsilon, \phi, x), \qquad \epsilon \sim p(\epsilon).} \tag{146}$$

For a factorized Gaussian,

$$\boxed{z = \mu + \sigma \odot \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I).} \tag{147}$$

For a full-covariance Gaussian,

$$\boxed{z = \mu + L\epsilon, \qquad \Sigma = LL^T.} \tag{148}$$

Finally, the direction of the KL divergence explains the mass-covering behavior:

$$\boxed{D_{\mathrm{KL}}(q\|p) = \mathbb{E}_q\left[\log \frac{q}{p}\right]} \tag{149}$$

which strongly penalizes the model for assigning insufficient probability to regions where the data distribution has probability.

Therefore, if an exact fit is impossible,

$$\boxed{\text{the learned generative distribution may become broader}} \tag{150}$$

and consequently may have a larger variance than the empirical data distribution.

22 Conclusion

Variational Autoencoders provide a principled framework for learning deep latent-variable generative models using variational inference and stochastic gradient optimization.

The encoder $q_\phi(z|x)$ approximates the otherwise intractable posterior $p_\theta(z|x)$. The central training objective is the ELBO,

$$\mathcal{L}_{\theta,\phi}(x) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x,z) - \log q_\phi(z|x)], \tag{151}$$

which is a lower bound on the marginal log-likelihood.

The gap between the ELBO and the marginal log-likelihood is exactly the KL divergence between the approximate and true posterior. Therefore, maximizing the ELBO simultaneously improves the generative model and the quality of the approximate posterior.

For continuous latent variables, the reparameterization trick transforms stochastic sampling into a differentiable deterministic operation with externally introduced noise. This makes the ELBO amenable to backpropagation and stochastic gradient descent.

Gaussian posterior distributions are particularly convenient because their reparameterizations have simple Jacobians. The factorized Gaussian uses

$$z = \mu + \sigma \odot \epsilon, \tag{152}$$

while a full-covariance Gaussian can use

$$z = \mu + L\epsilon, \qquad \Sigma = LL^T. \tag{153}$$

The KL decomposition also provides an important interpretation of VAE training. The joint KL between the data–encoder distribution and the generative model contains both a data-fitting component and a posterior-approximation component.

A major practical challenge is posterior collapse, where

$$q_\phi(z|x) \approx p(z). \tag{154}$$

In this situation, the latent variable contains little information about the input. KL annealing and free bits are two techniques that can mitigate this problem.

Finally, the direction of the KL divergence is important. The forward KL,

$$D_{\mathrm{KL}}(q\|p), \tag{155}$$

penalizes the generative model strongly when it fails to cover regions occupied by the data distribution. This mass-covering behavior explains why, when the model family cannot perfectly represent the data distribution, the learned distribution may become broader and have higher variance than the empirical data distribution.

Overall, the VAE framework can be understood as a coordinated optimization of three interconnected objectives:

$$\boxed{\begin{array}{rcl} \text{Learn a good generative model} & \Longleftrightarrow & \text{maximize likelihood,} \\ \text{Learn a good inference model} & \Longleftrightarrow & q_\phi(z|x) \approx p_\theta(z|x), \\ \text{Make both tractable} & \Longleftrightarrow & \text{optimize the ELBO using reparameterization.} \end{array}} \tag{156}$$

These ideas form the mathematical foundation for understanding more advanced variational methods, normalizing flows, importance-weighted autoencoders, and modern latent-variable generative models.